

Understanding Percentiles, Quartiles, IQR, and Z-Score in Machine Learning

Outline

1. Introduction
2. Percentiles and Quartiles
3. Interquartile Range (IQR)
4. Z-Score
5. Python Implementation & Outlier Detection

2. Percentiles and Quartiles

Definition:

"Percentile tells us the value below which a given percentage of observations fall."

Example:

"If your test score is in the 90th percentile, it means you scored better than 90% of the people."

Formula (Conceptually):

Percentile rank = (number of values below x / total number of values) × 100

Quartiles:

"Quartiles are special percentiles that divide data into four parts."

- **Q1 (25th percentile):** 25% of data lies below this value.
- **Q2 (50th percentile):** This is the **median**.
- **Q3 (75th percentile):** 75% of data lies below this value.

ML Connection:

"We often use percentiles to handle outliers in data.

For example, when preprocessing numerical features, we can remove values below the 1st percentile or above the 99th percentile to reduce the effect of extreme data points."

3. Interquartile Range (IQR)

Definition:

"IQR measures the spread of the middle 50% of data — it's calculated as Q3 minus Q1."

Formula:

$IQR = Q3 - Q1$

Interpretation:

"A smaller IQR means data points are tightly packed.

A larger IQR means data has high variability."

Outlier Detection Rule:

Any point below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is an **outlier**.

ML Connection:

"IQR-based filtering is common before training models to prevent extreme values from skewing results — for example, in **housing price prediction**, or **salary datasets**, where a few very large values can distort the model's understanding of normal behavior."

4. Z-Score

Definition:

"Z-score tells us how many standard deviations a value is away from the mean."

Formula:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- x = individual value
- μ = mean of the dataset
- σ = standard deviation

Interpretation:

- $z = 0 \rightarrow$ value is exactly at the mean
- $z = 1 \rightarrow$ 1 standard deviation above mean
- $z = -2 \rightarrow$ 2 standard deviations below mean

ML Connection:

“Z-score normalization is part of **standard scaling**, which transforms all features to have mean 0 and standard deviation 1.

This is critical for models like **KNN**, **SVM**, or **Gradient Descent-based models**, which are sensitive to feature scale.”

Summary

Concept	Formula	Use in ML
Percentile	position measure	Feature understanding
Quartile	Q1, Q2, Q3 (25th, 50th, 75th)	Spread summary
IQR	$Q3 - Q1$	Outlier detection
Z-score	$(x - \mu) / \sigma$	Standardization & scaling

Conclusion

To summarize:

- **Percentiles and quartiles** help describe data spread.
- **IQR** shows the range of central data and helps detect outliers.
- **Z-score** standardizes data and helps identify extreme values.

Together, these tools are fundamental for **data preprocessing**, **feature scaling**, and **robust ML model performance**.